

Data Structure Examples (Non-linear)

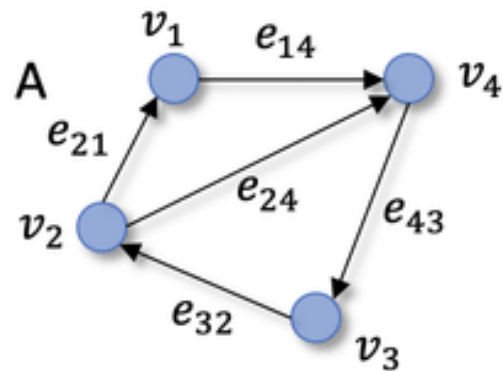
Graph

A generalization of the simple concept of a set of dots, links, edges or arcs.

Representation: Graph $G = (V, E)$ consists set of vertices denoted by V , or by $V(G)$ and set of edges E , or $E(G)$

Graph Representation – Adjacency Matrix

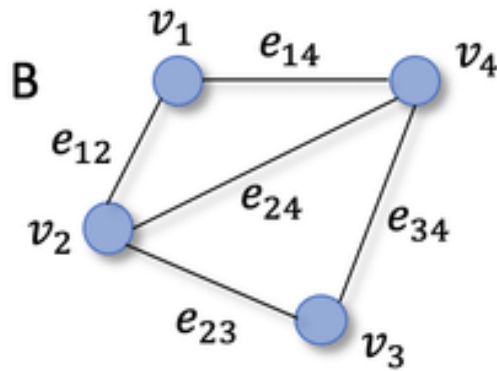
A Directed graph $G(V,E)$



E

	v_1	v_2	v_3	v_4
v_1	0	0	0	1
v_2	1	0	0	1
v_3	0	1	0	0
v_4	0	0	1	0

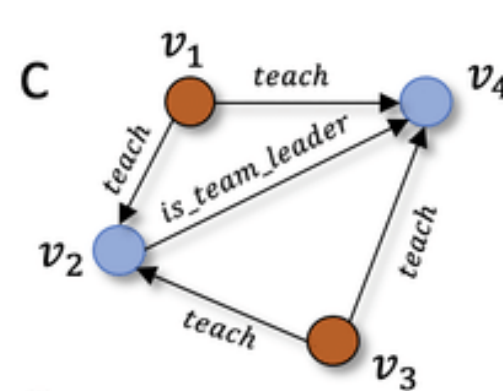
B Undirected graph $G(V,E)$



F

	v_1	v_2	v_3	v_4
v_1	0	1	0	1
v_2	1	0	1	1
v_3	0	1	0	1
v_4	1	1	1	0

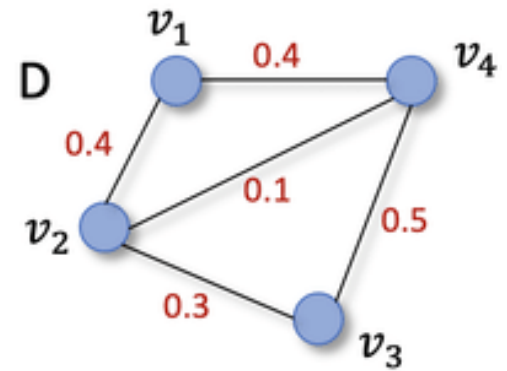
C Knowledge graph $G(V,E)$



G

	v_1	v_2	v_3	v_4
v_1	0	1	0	1
v_2	0	0	0	1
v_3	0	1	0	1
v_4	0	0	0	0

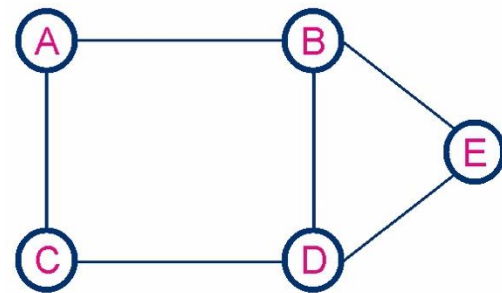
D Weighted graph $G(V,E)$



H

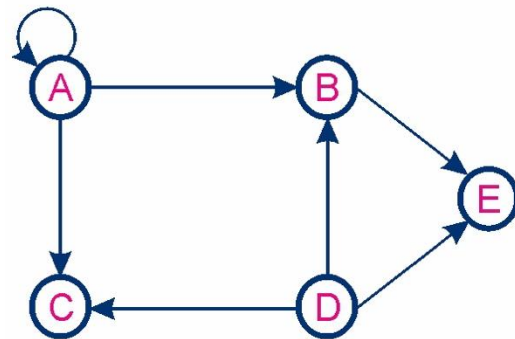
	v_1	v_2	v_3	v_4
v_1	0	0.4	0	0.4
v_2	0.4	0	0.3	0.1
v_3	0	0.3	0	0.5
v_4	0.4	0.1	0.5	0

Graph Representation – Adjacency List



Undirected Graph

A → B → C
B → A → D → E
C → A → D
D → B → C → E
E → B → D



Directed Graph

A → A → B → C
B → E
C →
D → B → C → E
E →